

SHETH MVLU COLLEGE

SUBJECT:- WSN

Aim: Performance Evaluation of MANET Protocols

Measure throughput, packet delivery ratio, and end-to-end delay for different MANET routing protocols.

CODE:

```
set ns [new Simulator]

set tracefd [open manet_aodv.tr w]
$ns trace-all $tracefd

set namfd [open manet_aodv.nam w]
$ns namtrace-all-wireless $namfd 800 800

set topo [new Topography]
$topo load_flatgrid 800 800

create-god 5

set chan_1_ [new Channel/WirelessChannel]

$ns node-config \
    -adhocRouting AODV \
    -llType LL \
    -macType Mac/802_11 \
    -ifqType Queue/DropTail/PriQueue \
    -ifqLen 50 \
    -antType Antenna/OmniAntenna \
    -propType Propagation/TwoRayGround \
    -phyType Phy/WirelessPhy \
    -channel $chan_1_ \
    -topoInstance $topo \
    -agentTrace ON \
    -routerTrace ON \
    -macTrace ON \
    -movementTrace ON

for {set i 0} {$i < 5} {incr i} {
    set node_($i)[ns node]
}

$node_(0) set X_ 100
$node_(0) set Y_ 100

$node_(1) set X_ 250
$node_(1) set Y_ 200

$node_(2) set X_ 400
$node_(2) set Y_ 300

$node_(3) set X_ 550
```

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```
$node_(3) set Y_ 400

$node_(4) set X_ 700
$node_(4) set Y_ 500

for {set i 0} {$i < 5} {incr i} {
    node_$i set Z_ 0
    $ns initial_node_pos node_$i 20
}

set udp0 [new Agent/UDP]
$ns attach-agent $node_(0) $udp0

set null0 [new Agent/Null]
$ns attach-agent $node_(4) $null0

$ns connect $udp0 $null0

set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set interval_ 0.1
$cbr0 attach-agent $udp0

nsat1.0"node_(1) setdest 300 300 10"
nsat2.0"node_(2) setdest 450 350 10"
nsat3.0"node_(3) setdest 600 450 10"

nsat1.0"cbr0 start"
nsat8.0"cbr0 stop"

proc finish {} {
    global ns tracefd namfd

    $ns flush-trace
    close $tracefd
    close $namfd

    exec nam manet_aodv.nam &
    exit 0
}

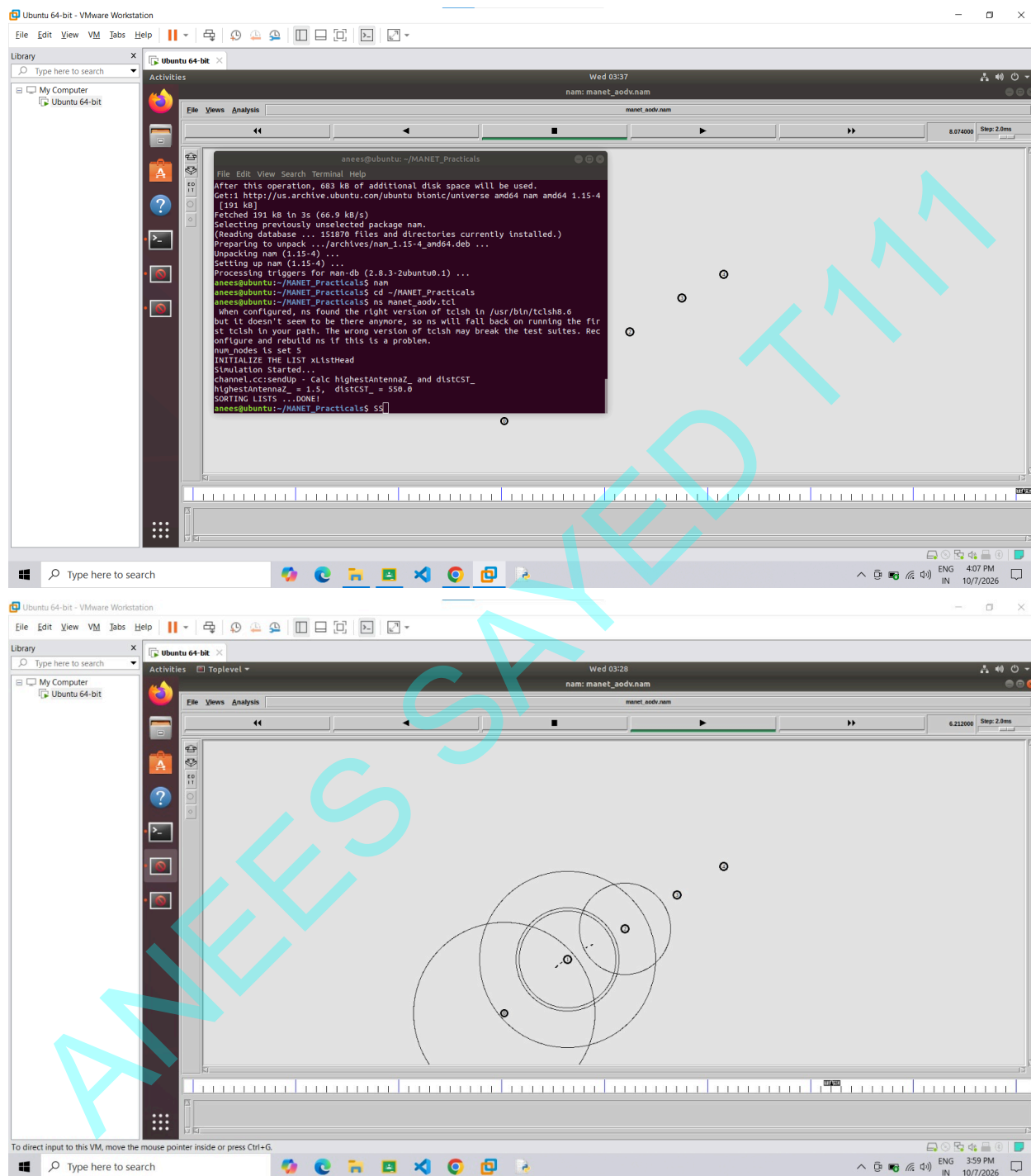
$ns at 10.0 "finish"

puts "Simulation Started..."
$ns run
```

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OUTPUT:



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Observation

- Five wireless nodes were created in the simulation area.
- Node 0 acted as the source and Node 4 as the destination.
- UDP was used as the transport protocol.
- CBR traffic generated continuous packet transmission.
- AODV established routes dynamically between nodes.
- Packet delivery was successfully visualized in NAM.

Result

The MANET network was successfully implemented using the AODV routing protocol. Data packets were transmitted from the source node to the destination node, and the communication process was observed through NAM animation.